

$$\begin{aligned}
& \frac{1}{16\pi^2} \left(\frac{1}{432} m_\Sigma^2 (s_\gamma^2 y_d^{11p} (5 g_1^2 \overline{y_d}^{12p} - 36 c_\gamma^2 \overline{y_d}^p \overline{y_d}^{12s} y_d^{rs}) + \right. \\
& s_\gamma^2 y_u^{11p} (-13 g_1^2 \overline{y_u}^{12p} + 36 s_\gamma^2 \overline{y_u}^p \overline{y_u}^{12s} y_u^{rs})) + \frac{1}{81} \sum_p g_1^4 \text{LF}_{3,0} [m_\Sigma^p] \delta_{11i2} - \\
& g_1^4 \text{LF}_{4,-1} [m_\Sigma^p] \delta_{11i2} + \frac{4}{405} \sum_p g_1^4 \text{LF}_{5,-2} [m_\Sigma^p] \delta_{11i2} + \frac{1}{27} \sum_p g_1^4 \text{LF}_{3,0} [m_\Sigma^p] \delta_{11i2} - \\
& g_1^4 \text{LF}_{4,-1} [m_\Sigma^p] \delta_{11i2} + \frac{4}{135} \sum_p g_1^4 \text{LF}_{5,-2} [m_\Sigma^p] \delta_{11i2} + \frac{1}{54} \sum_p g_1^4 \text{LF}_{3,0} [m_\Sigma^p] \delta_{11i2} - \\
& g_1^4 \text{LF}_{4,-1} [m_\Sigma^p] \delta_{11i2} + \frac{2}{135} \sum_p g_1^4 \text{LF}_{5,-2} [m_\Sigma^p] \delta_{11i2} + \\
& g_1^4 \text{LF}_{3,0} [m_\Sigma^p] \delta_{11i2} - \frac{2}{405} \sum_p g_1^4 \text{LF}_{4,-1} [m_\Sigma^p] \delta_{11i2} + \frac{2}{405} \sum_p g_1^4 \text{LF}_{5,-2} [m_\Sigma^p] \delta_{11i2} + \\
& g_1^4 \text{LF}_{3,0} [m_\Sigma^p] \delta_{11i2} - \frac{5}{54} \sum_p g_1^4 \text{LF}_{4,-1} [m_\Sigma^p] \delta_{11i2} + \frac{16}{405} \sum_p g_1^4 \text{LF}_{5,-2} [m_\Sigma^p] \delta_{11i2} + \\
& \left. 2 y_d^{11p} (g_1^2 \overline{y_d}^{12p} - 9 c_\gamma^2 \overline{y_d}^p \overline{y_d}^{12s} y_d^{rs}) + c_\gamma^2 y_u^{11p} (-2 g_1^2 \overline{y_u}^{12p} + 9 s_\gamma^2 \overline{y_u}^p \overline{y_u}^{12s} y_u^{rs}) \right) \\
& [m_\Sigma] + \frac{1}{24} g_1^2 (-s_\gamma^2 \overline{y_d}^{12p} y_d^{11p} + c_\gamma^2 \overline{y_u}^{12p} y_u^{11p}) \text{LF}_{2,1} [m_\Sigma] + \\
& 2 (9 s_\gamma^2 \overline{y_d}^{12p} y_d^{11p} - 9 c_\gamma^2 \overline{y_u}^{12p} y_u^{11p} + 4 g_1^2 \delta_{11i2}) \text{LF}_{3,0} [m_\Sigma] - \\
& 4 \text{LF}_{4,-1} [m_\Sigma] \delta_{11i2} + \frac{2}{135} g_1^4 \text{LF}_{5,-2} [m_\Sigma] \delta_{11i2} + \frac{1}{54} g_1^4 \text{LF}_{3,0} [\tilde{\mu}] \delta_{11i2} + \\
& \text{LF}_{4,-1} [\tilde{\mu}] \delta_{11i2} - \frac{4}{135} g_1^4 \text{LF}_{5,-2} [\tilde{\mu}] \delta_{11i2} + \frac{1}{18} g_1^2 c_\gamma^2 \overline{y_d}^{12p} y_d^{11p} \text{LF}_{2,1,0} [m_1, m_\Sigma^p] - \\
& c_\gamma^2 \overline{y_d}^{12p} y_d^{11p} \text{LF}_{3,1,-1} [m_1, m_\Sigma^p] + \frac{1}{18} g_1^2 c_\gamma^2 \overline{y_d}^{12p} y_d^{11p} \text{LF}_{4,-1,-2} [m_1, m_\Sigma^p] + \\
& 4 \text{LF}_{2,1,0} [m_1, m_\Sigma^{11}] \delta_{11i2} + \frac{1}{2592} g_1^4 \text{LF}_{2,2,-1} [m_1, m_\Sigma^{11}] \delta_{11i2} - \\
& 4 \text{LF}_{3,1,-1} [m_1, m_\Sigma^{11}] \delta_{11i2} + \frac{1}{2592} g_1^4 \text{LF}_{4,-1,-2} [m_1, m_\Sigma^{11}] \delta_{11i2} + \\
& 4 \text{LF}_{2,1,0} [m_1, m_\Sigma^{12}] \delta_{11i2} + \frac{1}{2592} g_1^4 \text{LF}_{2,2,-1} [m_1, m_\Sigma^{12}] \delta_{11i2} - \\
& 4 \text{LF}_{3,1,-1} [m_1, m_\Sigma^{12}] \delta_{11i2} + \frac{1}{2592} g_1^4 \text{LF}_{4,-1,-2} [m_1, m_\Sigma^{12}] \delta_{11i2} - \\
& s_\gamma^2 \overline{y_u}^{12p} y_u^{11p} \text{LF}_{2,1,0} [m_1, m_\Sigma^p] + \frac{1}{9} g_1^2 s_\gamma^2 \overline{y_u}^{12p} y_u^{11p} \text{LF}_{3,1,-1} [m_1, m_\Sigma^p] - \\
& s_\gamma^2 \overline{y_u}^{12p} y_u^{11p} \text{LF}_{4,1,-2} [m_1, m_\Sigma^p] + \frac{1}{96} g_1^2 g_2^2 \text{LF}_{2,1,0} [m_2, m_\Sigma^{11}] \delta_{11i2} + \\
& g_2^2 \text{LF}_{2,2,-1} [m_2, m_\Sigma^{11}] \delta_{11i2} - \frac{1}{48} g_1^2 g_2^2 \text{LF}_{3,1,-1} [m_2, m_\Sigma^{11}] \delta_{11i2} + \\
& g_2^2 \text{LF}_{4,1,-2} [m_2, m_\Sigma^{11}] \delta_{11i2} + \frac{1}{96} g_1^2 g_2^2 \text{LF}_{2,1,0} [m_2, m_\Sigma^{12}] \delta_{11i2} + \\
& g_2^2 \text{LF}_{2,2,-1} [m_2, m_\Sigma^{12}] \delta_{11i2} - \frac{1}{48} g_1^2 g_2^2 \text{LF}_{3,1,-1} [m_2, m_\Sigma^{12}] \delta_{11i2} + \\
& g_2^2 \text{LF}_{4,1,-2} [m_2, m_\Sigma^{12}] \delta_{11i2} + \frac{2}{3} g_3^2 c_\gamma^2 \overline{y_d}^{12p} y_d^{11p} \text{LF}_{2,1,0} [m_3, m_\Sigma^p] - \\
& c_\gamma^2 \overline{y_d}^{12p} y_d^{11p} \text{LF}_{3,1,-1} [m_3, m_\Sigma^p] + \frac{2}{3} g_3^2 c_\gamma^2 \overline{y_d}^{12p} y_d^{11p} \text{LF}_{4,1,-2} [m_3, m_\Sigma^p] + \\
& g_3^2 \text{LF}_{2,1,0} [m_3, m_\Sigma^{11}] \delta_{11i2} + \frac{1}{54} g_1^2 g_3^2 \text{LF}_{2,2,-1} [m_3, m_\Sigma^{11}] \delta_{11i2} - \\
& g_3^2 \text{LF}_{3,1,-1} [m_3, m_\Sigma^{11}] \delta_{11i2} + \frac{1}{54} g_1^2 g_3^2 \text{LF}_{4,-1,-2} [m_3, m_\Sigma^{11}] \delta_{11i2} + \\
& g_3^2 \text{LF}_{2,1,0} [m_3, m_\Sigma^{12}] \delta_{11i2} + \frac{1}{54} g_1^2 g_3^2 \text{LF}_{2,2,-1} [m_3, m_\Sigma^{12}] \delta_{11i2} - \\
& g_3^2 \text{LF}_{3,1,-1} [m_3, m_\Sigma^{12}] \delta_{11i2} + \frac{1}{54} g_1^2 g_3^2 \text{LF}_{4,-1,-2} [m_3, m_\Sigma^{12}] \delta_{11i2} - \\
& s_\gamma^2 \overline{y_u}^{12p} y_u^{11p} \text{LF}_{2,1,0} [m_3, m_\Sigma^p] + \frac{1}{3} g_3^2 s_\gamma^2 \overline{y_u}^{12p} y_u^{11p} \text{LF}_{3,1,-1} [m_3, m_\Sigma^p] - \\
& s_\gamma^2 \overline{y_u}^{12p} y_u^{11p} \text{LF}_{4,1,-2} [m_3, m_\Sigma^p] + \frac{1}{18} g_1^2 \overline{y_d}^{12p} y_d^{11p} \text{LF}_{2,1,0} [m_\Sigma^p, \tilde{\mu}] - \\
& \overline{y_d}^{12p} y_d^{11p} \text{LF}_{2,2,-1} [m_\Sigma^p, \tilde{\mu}] - \frac{1}{36} g_1^2 \overline{y_d}^{12p} y_d^{11p} \text{LF}_{3,1,-1} [m_\Sigma^p, \tilde{\mu}] + \\
& (c_\gamma \overline{a_d}^{\text{pr}} - s_\gamma \tilde{\mu} \overline{y_d}^{\text{pr}}) (c_\gamma a_d^{\text{pr}} - s_\gamma \tilde{\mu} y_d^{\text{pr}}) \text{LF}_{2,2,0} [m_\Sigma^r, m_\Sigma^p] \delta_{11i2} - \\
& (c_\gamma \overline{a_d}^{\text{pr}} - s_\gamma \tilde{\mu} \overline{y_d}^{\text{pr}}) (c_\gamma a_d^{\text{pr}} - s_\gamma \tilde{\mu} y_d^{\text{pr}}) \text{LF}_{3,2,-1} [m_\Sigma^r, m_\Sigma^p] \delta_{11i2} + \\
& (c_\gamma \overline{a_e}^{\text{pr}} - s_\gamma \tilde{\mu} \overline{y_e}^{\text{pr}}) (c_\gamma a_e^{\text{pr}} - s_\gamma \tilde{\mu} y_e^{\text{pr}}) \text{LF}_{2,2,0} [m_\Sigma^r, m_\Sigma^p] \delta_{11i2} - \\
& (c_\gamma \overline{a_e}^{\text{pr}} - s_\gamma \tilde{\mu} \overline{y_e}^{\text{pr}}) (c_\gamma a_e^{\text{pr}} - s_\gamma \tilde{\mu} y_e^{\text{pr}}) \text{LF}_{3,2,-1} [m_\Sigma^r, m_\Sigma^p] \delta_{11i2} - \\
& (c_\gamma \overline{a_e}^{\text{pr}} - s_\gamma \tilde{\mu} \overline{y_e}^{\text{pr}}) (c_\gamma a_e^{\text{pr}} - s_\gamma \tilde{\mu} y_e^{\text{pr}}) \text{LF}_{3,1,0} [m_\Sigma^r, m_\Sigma^p] \delta_{11i2} - \\
& (c_\gamma \overline{a_e}^{\text{pr}} - s_\gamma \tilde{\mu} \overline{y_e}^{\text{pr}}) (c_\gamma a_e^{\text{pr}} - s_\gamma \tilde{\mu} y_e^{\text{pr}}) \text{LF}_{3,2,-1} [m_\Sigma^r, m_\Sigma^p] \delta_{11i2} + \\
& (c_\gamma \overline{a_e}^{\text{pr}} - s_\gamma \tilde{\mu} \overline{y_e}^{\text{pr}}) (c_\gamma a_e^{\text{pr}} - s_\gamma \tilde{\mu} y_e^{\text{pr}}) \text{LF}_{4,1,-1} [m_\Sigma^r, m_\Sigma^p] \delta_{11i2} - \\
& (c_\gamma \overline{a_e}^{\text{pr}} - s_\gamma \tilde{\mu} \overline{y_e}^{\text{pr}}) (c_\gamma a_e^{\text{pr}} - s_\gamma \tilde{\mu} y_e^{\text{pr}}) \text{LF}_{5,1,-2} [m_\Sigma^r, m_\Sigma^p] \delta_{11i2} - \\
& (c_\gamma \overline{a_d}^{\text{pr}} - s_\gamma \tilde{\mu} \overline{y_d}^{\text{pr}}) (c_\gamma a_d^{\text{pr}} - s_\gamma \tilde{\mu} y_d^{\$$